True Consensus from Subtleties - Detecting Cognitive Synchrony in Two-person Interactions Based on Multimodal Physiological and Behavioral Signals

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Abstract

We study the problem of detecting whether consensus in a dyadic interaction reflects genuine information integration or a superficial agreement. While recent studies have linked interpersonal physiological synchrony to teamwork and decision quality, current multimedia datasets and models provide limited support for modeling consensus authenticity in conflict-driven interactions. To address this gap, we introduce CODEC (COnsensus DEtection in Conflict), a dyadic multimodal dataset collected with a hidden-profile paradigm that naturally induces disagreement, negotiation, and consensus formation. The dataset includes synchronized wearable physiological and behavioral streams from 13 dyads, together with fine-grained temporal annotations of interaction states. We also present a dynamics-informed late-fusion model that combines subject-aware prompt tuning, cross-modal attention guided by a cross-recurrence (CRQA) prior, and supervised contrastive learning to classify true versus false consensus from continuous multimodal segments. Under leave-one-group-out evaluation, the proposed model achieves a weighted F1 score of 0.85 and consistently improves over early-fusion and sequential baselines. Our results suggest that multimodal physiological-behavioral signals offer a promising direction for analyzing consensus quality beyond overt audiovisual behavior alone. The dataset can be accessed through https://github.com/CODEC-MM/Dataset.

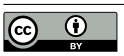
CCS Concepts

• **Human-centered computing** → **Collaborative and social computing**; • **Computing methodologies** → *Artificial intelligence*; • **Applied computing**;

Keywords

Multimodal Fusion; Affective Computing; Social Signal Processing; Multimodal Analysis; Interpersonal Physiological Synchrony; Consensus Detection

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1 Introduction

Effective group decision-making is the cornerstone of modern organization operation and human social collaboration, especially in high-risk areas such as medical consultation, airline crew collaboration and social governance [9, 14, 29]. The quality of decision-making highly depends on whether the team members can achieve true information integration and cognitive unity. However, the decision-making process is highly vulnerable to the **Groupthink** phenomenon [26, 35]. In this state, in order to maintain superficial social harmony, avoid interpersonal conflict, or simply out of escape from complex information processing, participants often choose to hide their dissent and compromise unilaterally under intense social pressure or cognitive fatigue, thus forming a **False Consensus** that is not verified by strict logic [18, 26]. From the perspective of traditional visual and acoustic analysis, the False Consensus is extremely similar to the True Consensus in appearance. For example, both parties may be nodding frequently, smiling courteously, or maintaining submissive silences. Such highly subjective controllability together with social camouflage constitutes a significant detection blind area for current multimodal interaction systems when assessing the quality of deep collaboration [15, 22]. If an AI-assisted system cannot penetrate this veneer based on social etiquette, it cannot provide timely intervention and information supplement when the quality of collaboration deteriorates.

In order to overcome the inherent limitations of traditional audiovisual modalities in detecting higher-order collaborative cognitive states, physiological computing provides an objective window to directly observe human internal emotional fluctuations and cognitive load [13]. Physiological signals innervated by the autonomic nervous system, such as galvanic skin activity and heart rate variability, which reflect sympathetic arousal, have biological properties that are extremely difficult to subjectively disguise [10, 27]. Traditional physiological acquisition devices are often extremely bulky and interfere with natural interaction. However, in recent years, the emergence of emerging research-grade wearable devices, such as

Emotibit, not only provides subjects with insensitive wearing comfort and research-level data accuracy, but also makes it possible to automatically synchronize multi-source heterogeneous signals because of their hardware-level characteristics [7, 20]. This provides us with a perfect prerequisite to capture microscopic physiological changes in natural social scenarios.

In recent years, the intersection of psychology and social neuroscience has proposed that when individuals deeply resonate at the cognitive level or collaborate to solve complex problems, their interpersonal physiological synchrony will show specific temporal coupling and state transition patterns [4, 19]. Inspired by this, this paper proposes a core computational assumption: When people experience real cognitive conflict and finally reach a consensus through logical persuasion, the physiological level will experience a dynamic transition from high pressure confrontation to pressure release, which provides a candidate physiological marker for distinguishing True and False Consensus. By introducing a deep learning network with powerful sequence modeling ability, these microscopic physiological temporal characteristics are jointly modeled. This enables decoding of internal micro cognitive state under the controlled conditions studied here.

Although physiological synchrony has shown great potential for assessing binary interactions, the current multimodal machine learning field still lacks high-quality continuous interaction datasets focusing on the cognitive states of microscopic decision-making. Most existing emotion datasets induce single, discrete internal emotions by passively watching emotive arousal videos or listening to music, which seriously lacks high-intensity cognitive games in real social interaction scenarios [8, 10]. In addition, multimodal feature-level fusion algorithms for two-person continuous physiological signals are still in the exploratory stage. How to effectively deal with the heterogeneity, time delay and high individual differences between different physiological modalities is a major technical problem in multimodal representation learning [10].

Based on the above challenges and hardware capabilities, this study proposes a new topic of **multimodal cognitive synchrony detection**, and constructs a complete multimodal framework including two-person interaction data collection, micro-state cross-labeling, nonlinear dynamics analysis, and spatio-temporal deep learning fusion. The core contributions of this study can be summarized in three dimensions:

1) Firstly, we proposed the task of Multimodal Cognitive Synchrony Detection and construct CODEC, a dyadic multimodal physiological interaction dataset specifically designed for fine-grained consensus detection. By adapting the **hidden profile paradigm**, we successfully reproduce the complex decision-making process in a laboratory environment with sub-second microstate labels, ensuring data quality through rigorous double-blind consistency checks and calibrated physiological gradients via a clap-sync mechanism.

2) Secondly, this study introduces cross-recurrence quantification analysis (CRQA) into multimodal affective computing. Within CODEC, we quantify **True Consensus** as a structurally recurrent state accompanied by physiological stress release, whereas **False Consensus** is associated with markedly lower physiological arousal and reduced cognitive engagement.

3) Finally, to address the heterogeneity of dyadic physiological characteristics and feature entanglement, we propose a dynamics-informed spatio-temporal multimodal late fusion framework named DyadicSync-Net. By integrating Subject-Aware Prompt Tuning and Supervised Contrastive learning, and driven by dynamics-informed cross-modal attention, the network effectively fuses emotional and behavioral features. It achieves a competitive weighted F1 score of 0.85 in the continuous slice classification task of detecting authenticity consensus, outperforming traditional single-modal and early-fusion machine learning baselines, demonstrating the feasibility of penetrating pretext and diagnosing consensus through implicit physiological signals.

2 Related Work

2.1 Interpersonal Physiological Synchrony in Affective Computing

Multimodal affective computing fuses multi-source information to infer complex mental states [3, 42]. Moving beyond early audio-visual discrete emotion classification [31], research now targets social cognitive states in high-risk scenarios [28]. Thus, continuous physiological signals (e.g., EDA, EEG) have become vital data sources [11, 23, 25]. Traditional post-hoc group dynamic assessments suffer from lag. With advanced wearables, interpersonal physiological synchrony (IPS) provides an objective, real-time collaboration measure [4]. While higher IPS often predicts better performance [21], it can manifest negatively during intense cognitive friction (e.g., stress inducing defensive inhibition) . Currently, offline statistical analyses of IPS fall short for real-time machine learning predictions [26].

2.2 Cognitive Conflict Datasets and the Hidden Profile Paradigm

Mainstream datasets (e.g., SEED, DEAP) primarily focus on single-subject passive emotion induction, lacking the high-frequency cognitive evolution of real-world dyadic interactions [17]. Similarly, popular multimodal interaction datasets (e.g., IEMOCAP or MELD) predominantly provide utterance-level static labels, omitting the continuous temporal micro-state annotations needed to capture instantaneous debates [5, 24].

To address this, recent studies utilize the hidden profile paradigm (HPP) to assess macroscopic group dynamics [4, 40]. HPP allocates information asymmetrically; relying on shared information mimics Groupthink, whereas disclosing unique information through conflict enables True Consensus [1, 33]. While IPS under HPP predicts group information processing [26], prior works focus on retrospective, macroscopic statistical correlations. We diverge by extending HPP into multimodal deep learning as a **cognitive conflict engine** for fine-grained classification, thereby enriching our CODEC dataset with authentic interactions and granting our model stronger causal explanatory power.

2.3 Multimodal Temporal Deep Fusion Architectures

For physiological representation, deep networks (e.g., GCNs, Transformers) automatically extract high-order dynamic features [6, 12,

32]. Multimodal fusion is crucial for mining complementary information [2]. However, non-stationary physiological signals and motion artifacts pose challenges [25, 41]. Due to differing dynamic ranges (e.g., slow EDA vs. abrupt IMU), Early Fusion often causes feature contamination [2, 39].

Conversely, Late Fusion processes modalities independently before aggregation, offering superior noise resistance for heterogeneous data [26, 30]. Recently, Transformers set benchmarks in multimodal analysis by dynamically weighing features via self- and cross-attention [34, 36]. Inspired by this, we propose a two-stream attention Late Fusion Transformer tailored to dyadic interactions to advance the recognition of complex social cognitive states.

3 The CODEC Dataset and Problem Formulation

To formalize the new task of Multimodal Cognitive Synchrony Detection and overcome the limitations of existing datasets—which primarily focus on single-subject passive emotion induction and ignore masked social intentions—we constructed a specialized experimental pipeline to create the CODEC (COnsensus DEtection in Conflict) dataset. Rather than merely serving as an evaluation benchmark, the construction of this dataset functions as a highly controlled social signal elicitation engine, designed in strict accordance with experimental psychology protocols to capture authentic, conflict-driven dyadic interactions.

3.1 Participants and Hardware Deployment

We recruited a moderate sample size of 26 participants (including undergraduates and graduate students) and randomly paired them in pairs to form 13 two-person interaction groups, achieving the standard sample size for this type of research (DREAMER, 23 [16]; DEAP, 32 [17]). Note that hardware synchronization between independent wearable devices is a critical factor in this type of research. All experiments were conducted in a strictly controlled, soundproof laboratory environment.

Furthermore, this dataset is temporally dense rather than sparse in terms of the number of trials—each pair of participants engaged in continuous discussion for 15–30 minutes, and within a single experiment, the *conflict–consensus* cycle recurred multiple times as arguments were raised, debated, and resolved. Thousands of multimodal micro-clips can be generated within a time window of tens of seconds.

To capture implicit physiological signals with minimal interference to the subjects' natural body language, each participant was equipped with a research-grade wearable biosensor, EmotiBit (shown in Fig. 1), securely attached to the fingers of their non-dominant hand. The device synchronously recorded Electrodermal Activity (EDA) and Heart Rate (HR) to reflect sympathetic arousal, alongside 9-axis Inertial Measurement Unit (IMU) data (including a 3-axis accelerometer) at 25 Hz to quantify physical engagement and non-verbal communication intensity. Concurrently, a 4K ultra-HD camera and a high-fidelity array microphone system were deployed to capture wide-angle audiovisual signals, serving as the absolute ground truth for subsequent semantic annotation.

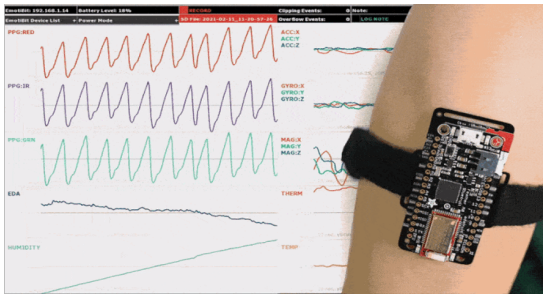


Figure 1: EmotiBit Biosignal System Deployment [7]

3.2 Cognitive Conflict Elicitation via HPP

Given the scarcity of high-frequency supervisory data for micro-level decision states, we departed from traditional passive emotion elicitation (e.g., video watching) and employed the classic Hidden Profile Paradigm (HPP) from social psychology. This paradigm deliberately creates a contradiction between individuals' initial preferences based on local information and the dyad's optimal solution based on global information.

Participants were instructed to act as a **Teacher Evaluation Committee** tasked with selecting the sole **Teacher of the Year** from three candidates through face-to-face deliberation. The profiles of the candidates were meticulously partitioned into **shared information** (available to both parties) and **unshared information** (available to only one party). The specific elicitation mechanisms are detailed in Table 1. Through this sophisticated architectural design of information, if subjects engage in superficial communication merely to expedite the experiment, they inevitably generate a False Consensus characterized by low cognitive engagement. Conversely, only by actively synthesizing unshared information and overcoming cognitive friction can they uncover the hidden optimal solution, thereby exhibiting the complete trajectory of reaching a True Consensus.

3.3 Modality Alignment via Clap-Sync Protocol

Aiming at the inherently non-stationary characteristics of physiological time series, we designed a rigorous preprocessing pipeline. A critical challenge within this pipeline is multimodal sensor fusion, where feature misalignment caused by minor clock drifts among independent devices can acutely degrade model performance. To overcome inherent latency and clock disparities, we integrated a physical "Clap-Sync" protocol into the experimental workflow.

Before initiating the formal debate phase, subjects were instructed to perform a synchronous clap. This crisp action produced not only an extreme impulse in the audio track but also a pronounced maximum peak in the 3D acceleration magnitude sequence of both IMUs. Using the video frame corresponding to this spatial extremum as the global baseline zero point, our alignment algorithm achieved strict sub-second anchoring of the dual-subject, four-channel high-frequency physiological signals with the video semantic sequences. After statistical and kinematic features were extracted over a dense 25 Hz sliding window, all digital signals were smoothed and uniformly aggregated after anti-aliasing filtering, ensuring complete retention of micro-movements. To prevent the sympathetic surge caused by the physical claps from contaminating the interactive

Table 1: Information Distribution and Elicitation Mechanisms of the Hidden Profile Task

Candidate	Info Type	Recipient	Information Details	Psychological Mechanism & Expected State
Candidate 1 (Dr. Miller)	Shared Info	P1, P2	Highly rated on student forums; classes feel like talk shows; extremely lenient grading.	**Decoy Option.** Without deep discussion, dyads easily reach a superficial agreement based on shared info, falling into a False Consensus.
	Unshared Info	P1 (Academic)	Frequently skips classes for commercial book signings, violating the teaching contract.	Potential trigger for cognitive conflict, requiring P1 to actively break the false harmony for disclosure.
	Unshared Info	P2 (Career)	Alumni surveys indicate graduates frequently fail technical interviews due to a lack of foundational knowledge.	Same as above.
Candidate 2 (Dr. Chen)	Shared Info	P1, P2	Students complain about massive workload; perceived as unsympathetic; induces high stress.	**Optimal Option.** Superficially unappealing, typically leading to strong initial rejection by subjects.
	Unshared Info	P1 (Academic)	Introduced cutting-edge LLM textbooks; grades assignments on weekends, showing high responsibility.	Necessitates intense debate. Overcoming this cognitive barrier leads to True Consensus.
	Unshared Info	P2 (Career)	Recruiters from top firms explicitly state that only this teacher's students can handle high-pressure jobs.	Core supporting argument, forcing deep information integration and value re-evaluation.

physiological baseline, a mandatory 120-second silent resting period was strictly enforced after the clap-sync.

3.4 Coarse-to-Fine Cross-Annotation

To map complex social interactions into high-quality label sequences supervisable by deep learning models, we developed a two-stage **coarse-to-fine** cross-annotation strategy.

Phase 1: Coarse-Grained Semantic Annotation. Two independent graduate-level annotators, professionally trained in organizational behavior and micro-expression analysis, reviewed the video recordings. They divided the continuous timeline into clear, non-overlapping interaction phases (e.g., Intense Conflict, True Consensus) based on verbal logic, intonation shifts, and micro-body movements (e.g., leaning forward to apply pressure, or defensive backward postures). The Cohen's Kappa coefficient between annotators reached 0.78, robustly confirming a high degree of subjective rater reliability [37]. These cues were used to segment the timeline; the remaining determinations were based on objective results identified from the hidden profile and cross-checked through structured post-task recall interviews with each participant. If the subject of the agreement was a decoy candidate, or if agreement was reached before any unshared items were revealed, the interaction was dropped regardless of how cooperative it appeared.

Phase 2: Fine-Grained Physiological Calibration. Human semantic annotation suffers from temporal imprecision, as physiological arousal typically exhibits a biological latency of 1 to 3 seconds. To compensate for this, we introduced an automatic fine-grained physiological boundary calibration algorithm. Within a ±5-second neighborhood of the identified semantic windows, the algorithm computed the first-order temporal derivative of the EDA sequence. It automatically located local maximum gradients (marking the absolute physical onset of emotional bursts) and local minimum gradients (marking stress release post-confrontation). This dual-stage methodology effectively transforms the macroscopic social behaviors into precise microscopic state boundaries. In practice the calibration consistently moves boundaries earlier, by 2.0–3.8 s across our logs, confirming that annotators mark an event only after the physiological onset they are reacting to. An Intense Conflict segment marked at 77.4 s, for instance, resolves to a 75.3–82.3 s window anchored on the EDA surge, while the True Consensus that follows is anchored on the steep arousal decline marking stress release.

4 Proposed Deep Learning Model

Having established the CODEC dataset and successfully captured the fine-grained physiological manifestations of dyadic interactions, the next critical challenge is computationally modeling these highly dynamic sequences. To this end, we propose DyadicSync-Net, a computational framework that deeply couples physiological dynamics with deep representation learning, as shown in Fig. 2. The framework is designed to address three major challenges inherent in our collected data and current affective computing: (1) individual heterogeneity in physiological baselines, (2) the lack of structural physical priors in cross-modal fusion, and (3) the severe representation collision between genuine and fabricated social states.

4.1 Subject-Aware Prompt Tuning for Physiological Heterogeneity

Electrodermal Activity (EDA) signals are susceptible to individual differences such as baseline skin impedance, metabolism, and

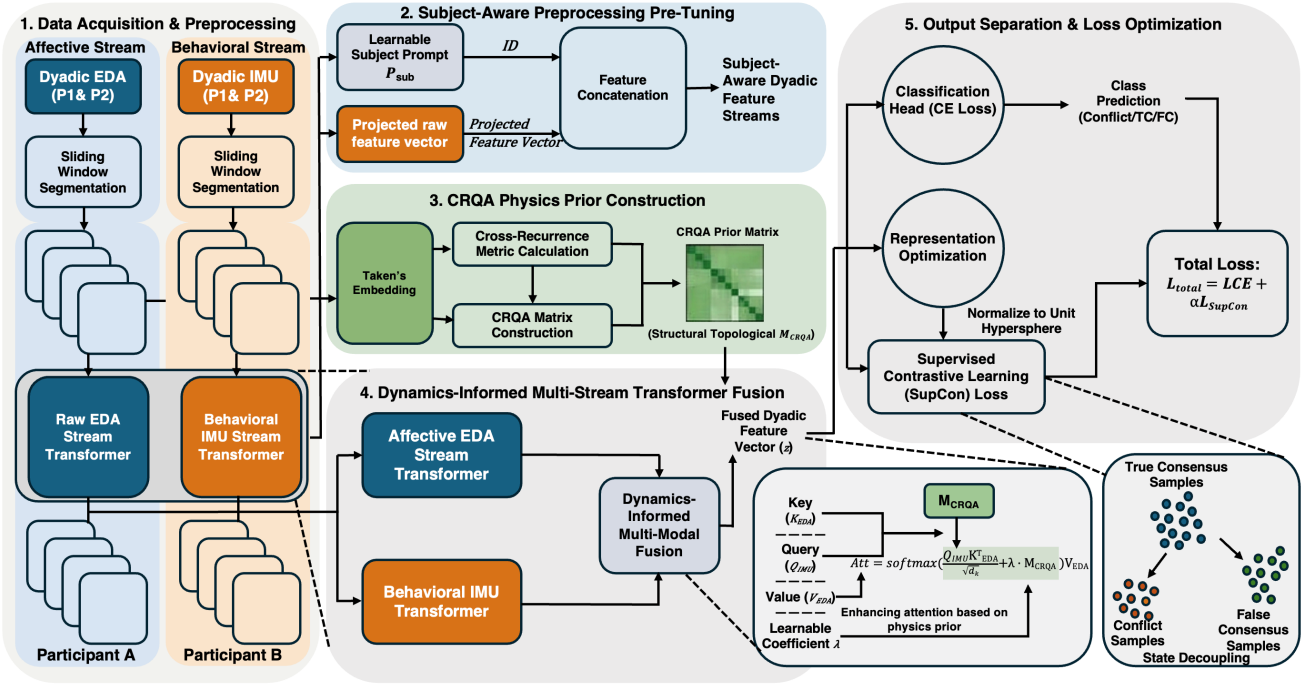


Figure 2: Proposed Dyadic Cognitive Synchrony Detection Framework

varying response thresholds. Forcing a unified network to process heterogeneous raw signals often leads to suboptimal generalization.

Inspired by prompt tuning mechanisms in large language models, we propose a Subject-Aware Prompt Tuning strategy. Before temporal encoding, we assign a learnable parameter matrix, termed the Subject-Specific Prompt $P_{sub} \in \mathbb{R}^{l_p \times d_{model}}$, to each participant. This prompt acts as an inherent physiological identity embedding. The prompt is prepended to the linearly projected continuous physiological sequence $X_{EDA} \in \mathbb{R}^{T \times d_{model}}$:

$$\tilde{X}^{(i)}_{EDA} = [P^{(i)}_{sub} \parallel X^{(i)}_{EDA}] \in \mathbb{R}^{(l_p+T) \times d_{model}} \quad (1)$$

where i denotes the subject ID and $\parallel$ represents the concatenation operation over the sequence length dimension. Through backpropagation, $P^{(i)}_{sub}$ adaptively learns the baseline distribution mapping of the specific user, effectively aligning heterogeneous multimodal streams into a unified feature space without requiring massive subject-specific network branches.

For the unseen dyad in the experiment, $P^{(i)}_{sub}$ is cold-started from the participant's pre-discussion resting baseline; after being processed through the frozen projection layer, it is filled into the l_p prompt slot via mean pooling. This process does not involve labels, samples from the discussion phase, or gradients from the test fold.

4.2 Non-linear Dynamics Prior via CRQA

Traditional linear statistical features fail to capture the complex, non-stationary interactions between two individuals. To extract the macroscopic structural state of the dyadic system, we introduce Cross-Recurrence Quantification Analysis (CRQA) to construct a topological dynamics prior [38].

Based on Takens' embedding theorem, the 1D physiological time series of the two subjects (denoted as u_A and u_B) are reconstructed into high-dimensional phase-space trajectory vectors [43]. Specifically, a state vector for Subject A at time i is formulated as $\vec{x}_i = [u_A(i), u_A(i+d), \ldots, u_A(i+(m-1)d)]$, where m is the embedding dimension and d is the time delay. The counterpart $\vec{y}_j$ is similarly constructed for Subject B. We define a cross-recurrence matrix $M_{CRQA} \in \mathbb{R}^{T \times T}$ to identify synchronized state recurrences:

$$M_{CRQA}(i,j) = \Theta(\varepsilon - ||\vec{x}_i - \vec{y}_j||) \quad (2)$$

where ε is the distance threshold and Θ is the Heaviside step function. This matrix explicitly maps the structural stability of the interpersonal physiological resonance, which serves as a crucial inductive bias for the subsequent deep fusion layer.

4.3 Dynamics-Informed Attention Fusion

Effectively fusing heterogeneous physiological (EDA) and behavioral (IMU) signals is a major technical obstacle. Given the high-frequency motion artifacts inherent in IMU data, early fusion strategies inevitably lead to cross-contamination of feature spaces. We design a parallel dual-stream Transformer architecture with a deeply customized cross-modal attention mechanism.

To mathematically suppress physiological pseudo-peaks caused by explicit physical movements, we use the behavioral high-order features (H_{IMU}) as the Query matrix, and the affective features (H_{EDA}) as the Key and Value matrices. Crucially, we inject the previously calculated structural matrix M_{CRQA} into the attention

formulation as the dynamics prior:

$$\text{Att}(Q_{IMU}, K_{EDA}, V_{EDA}) = \text{softmax}\left(\frac{Q_{IMU}K_{EDA}^T}{\sqrt{d_k}} + \lambda \cdot M_{CRQA}\right)V_{EDA} \tag{3}$$

where d_k is the scaling factor and λ is a learnable coefficient controlling the influence of the dynamics prior. This mechanism ensures that the network does not learn attention weights blindly from scratch. Instead, it is explicitly guided by the measured recurrence structure of interpersonal physiological coupling. When severe physical tremor is detected, the behavioral query effectively downregulates the fusion weight of the affective stream, unless sustained by a strong structural resonance indicated by M_{CRQA}.

4.4 State Decoupling via Supervised Contrastive Learning

A core difficulty in evaluating collaboration quality is that True Consensus and False Consensus appear remarkably similar on the surface (e.g., nodding, polite silence). Relying solely on a traditional Cross-Entropy (CE) loss often results in detrimental feature overlap in the latent space.

To achieve fine-grained state decoupling, we introduce a Multimodal Supervised Contrastive Learning (SupCon) loss. Before the final linear classifier, the fused dyadic representation vector z is projected into a normalized unit hypersphere. The SupCon loss explicitly forces the representations of slices from the same cognitive state (e.g., all True Consensus samples) to pull closer together, while pushing apart those from different states:

$$\mathcal{L}_{SupCon} = \frac{1}{|I|}\sum_{i\in I}\frac{-1}{|P(i)|}\sum_{p\in P(i)}\log\frac{\exp(z_i \cdot z_p/\tau)}{\sum_{a\in A(i)}\exp(z_i \cdot z_a/\tau)} \tag{4}$$

where I is the set of all indices in the mini-batch, i is the anchor sample, $P(i)$ represents the set of positive samples sharing the same label, $A(i)$ comprises all other samples in the mini-batch, and τ is the temperature hyperparameter. The final network is optimized end-to-end using a joint loss function:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \alpha\mathcal{L}_{SupCon} \tag{5}$$

where α balances the classification accuracy with the structural discriminability of the latent manifold.

5 Experiments and Results

5.1 Experimental Setup

Given the continuous multimodal sensor streams, we formulated the cognitive synchrony detection as a fine-grained multivariate time-series classification problem utilizing a sliding window strategy (window size: 10s, step: 2s). To reflect authentic social interactions, the dataset is collected for a considerable duration and preserves its imbalanced naturalistic distribution. All experiments were conducted on two NVIDIA P100 GPUs using the PyTorch framework. The proposed network was optimized utilizing the AdamW optimizer with a cosine annealing learning rate scheduler (peak LR: 1×10^{-4}, weight decay: 1×10^{-4}) for 30 epochs. To prevent over-fitting, a dropout rate of 0.1 was applied across all attention blocks, and the mini-batch size was set to 16.

5.2 Baseline Comparison for Cognitive Synchrony Detection

To rigorously evaluate the model's cross-domain generalization capability on unseen interactive dyads and mitigate the risk of overfitting on a limited data manifold, we executed a Leave-One-Group-Out (LOGO) cross-validation protocol, testing purely on unseen subjects. We comprehensively compared our proposed framework against five representative baselines and state-of-the-art models. To provide a holistic evaluation, especially considering the challenging naturalistic data distribution, we report Accuracy, Macro F1, True Consensus F1, and Weighted F1 scores.

Table 2: Performance Comparison of Multi-Modal Architectures under LOGO Cross-Validation

Model Architecture	Fusion Strategy	Accuracy	Macro F1	F1 (True Con.)	Weighted F1
SVM	Early Manual Concat	73.2%	0.48	0.51	0.69
Random Forest (RF)	Early Manual Concat	75.8%	0.52	0.54	0.72
Bi-LSTM	Early Sequential	80.3%	0.59	0.61	0.77
Transformer Encoder	Early Sequential	82.1%	0.64	0.65	0.79
Bi-LSTM	Late Dual-Stream	83.4%	0.68	0.69	0.81
Proposed Framework	**Cross-Modal Late**	**86.5%**	**0.76**	**0.78**	**0.85**

As summarized in Table 2, deep temporal models excellently outperform traditional static machine learning algorithms, emphasizing the necessity of capturing long-range spatio-temporal dynamics. Furthermore, late-fusion strategies consistently surpass their early-fusion counterparts, proving that decoupling heterogeneous modalities before aggregation effectively prevents high-frequency motion artifacts from contaminating subtle physiological variations. Our proposed framework achieves state-of-the-art performance across all metrics. Notably, it records a notable F1 score of 0.78 on the highly challenging True Consensus class, demonstrating its superior capability in capturing fleeting, high-order cognitive synchrony in naturalistic environments.

5.3 Ablation Study of Core Components

To quantitatively analyze the driving forces behind our framework's success, we conducted a step-by-step ablation study. We specifically highlight the improvements in Macro F1 and True Consensus F1, as these metrics best reflect the ability of the model to discriminate complex minority states in extreme data imbalance.

Table 3: Ablation Study on the Proposed Core Components

Architecture Configuration	Accuracy	Macro F1	F1 (True Con.)
Base Late Fusion Transformer	83.2%	0.66	0.68
+ Subject-Aware Prompt Tuning	84.4%	0.68	0.70
+ Dynamics Prior (CRQA)	85.1%	0.71	0.74
+ SupCon Loss (Full Model)	**86.5%**	**0.76**	**0.78**

As shown in Table 3, the performance trajectories exhibit a compelling and interpretable non-linear growth pattern, aligning perfectly with the theoretical design of each component:

Subject-Aware Prompt Tuning yields the most significant initial boost to global *Accuracy* (83.2% $\rightarrow$ 84.4%). By harmonizing the physiological baseline drift across varying individuals, it effectively aligns the majority class representations, though its impact on fine-grained cognitive state differentiation remains moderate.

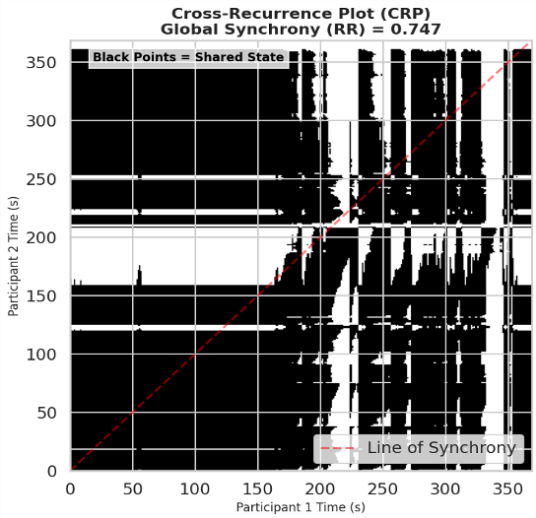


Figure 3: Cross-Recurrence Plot (CRP) of dyadic EDA. The high Global Synchrony ($RR = 0.747$) and prominent block structures (Laminarity) during True Consensus periods provide a robust nonlinear feature prior for the network.

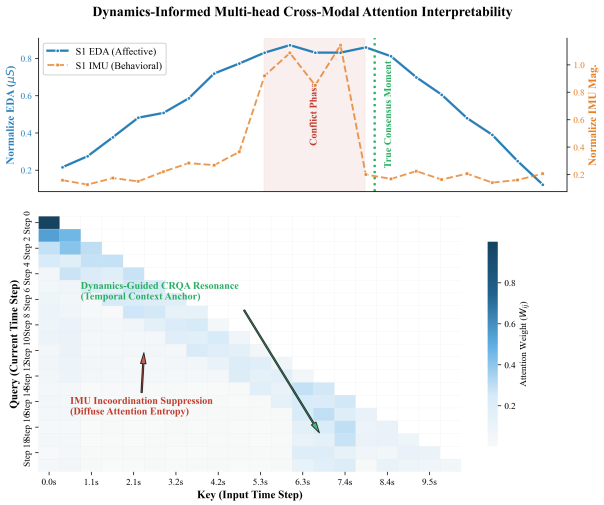


Figure 4: Visualization of dynamics-informed multi-head cross-modal attention. The network successfully suppresses IMU motion artifacts while anchoring to consensus moments via the CRQA-informed prior.

CRQA Dynamics Prior introduces a distinct paradigm shift. While its contribution to overall Accuracy is relatively marginal (+0.7%), it triggers a substantial surge in *True Consensus F1* (0.70 → 0.74). This empirical asymmetry proves that the topological constraint successfully prevents the cross-modal attention mechanism from being misled by spurious motion artifacts during intense physical engagement, precisely capturing genuine structural resonance.

Supervised Contrastive Loss serves as the critical final catalyst. By explicitly optimizing the inter-class margins in the latent manifold, it resolves the severe representation entanglement between explicit social masking (False Consensus) and genuine alignment. This leads to a profound leap in *Macro F1* (+0.05), pushing the global robustness and minority-class sensitivity to state-of-the-art levels.

5.4 Quantitative Feasibility Analysis

Before supervising the deep neural networks, we performed a rigorous quantitative feasibility study to ensure that the extracted physiological landmarks are statistically discriminative for dyadic cognitive states.

5.4.1 Statistical Validation of Interaction Landmarks. The statistical profiles across interaction states reveal distinct physiological signatures (Table 4).

Table 4: Physiological Metrics Across Interaction States (Mean Values)

Interaction State	Duration (s)	Synchrony (r)	Avg_Arousal (Z)	Avg_Motion (Z)
Conflict	72	-0.413	-0.125	0.092
Discussion	18	-0.314	-0.026	0.339
False Consensus	3	0.000	-0.780	0.102
True Consensus	6	0.000	-0.375	-0.152
Final Consensus	9	-0.449	-0.287	0.869

A primary observation is the arousal differentiation between adversarial and collaborative phases. Conflict states elicit considerably higher physiological arousal ($Z = -0.125$) compared to True Consensus ($Z = -0.375$), confirming that the transition to a unified cognitive state is accompanied by a measurable release of physiological stress.

Furthermore, the synchrony dynamics exhibit a shift from physiological divergence during conflict ($r = -0.413$) to neutralization during consensus ($r \approx 0$). This transition from negative synchrony (representing opposing arousal patterns) to a stable, neutral state suggests that the dyad has overcome initial cognitive friction to achieve a structural equilibrium.

5.4.2 Physiological Decoding of Consensus Authenticity. A core contribution of this feasibility analysis is the identification of the Physiological Signature of Agreement that distinguishes between true and False Consensus. Despite their behavioral similarities, these states are physiologically distinct:

False Consensus: Characterized by the lowest recorded arousal levels ($Z = -0.780$) and negligible physiological coherence. The short duration (< 5s) and lack of preceding stress-release patterns indicate cognitive disengagement and a Groupthink phenomenon where participants comply without active information processing.

True Consensus: Manifests as a moderate relaxation phase following intense cognitive interplay. The structural stability of this state is further confirmed by Cross-Recurrence Quantification Analysis (CRQA), where the real consensus phase exhibits extremely high global synchrony ($RR = 0.747$).

These findings empirically demonstrate the feasibility of using multimodal wearables to penetrate social masks and objectively assess the underlying quality of group decisions.

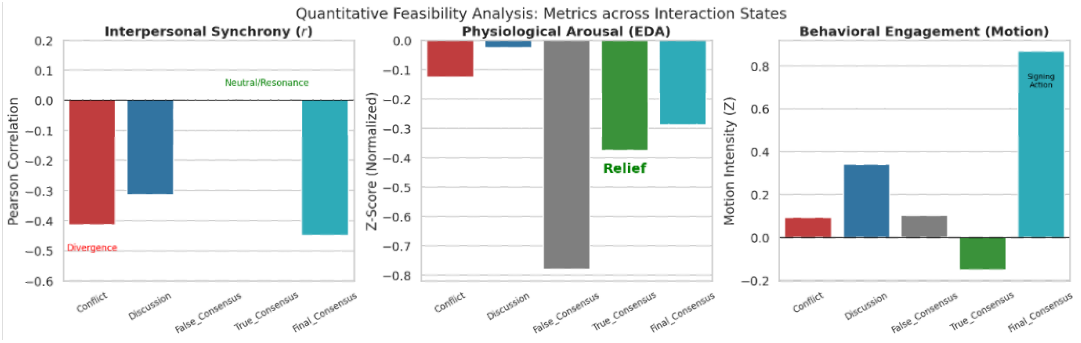


Figure 5: Quantitative Feasibility Analysis. Statistical distributions of interpersonal synchrony, physiological arousal (EDA), and behavioral engagement across five interaction micro-states, highlighting the distinct multimodal signatures of cognitive conflict and consensus.

5.5 Dynamics Visualization and Model Interpretability

To visually interpret the physiological foundation of our model's decisions, we employ Cross-Recurrence Plot (CRP) to analyze the nonlinear coupling between dyadic EDA streams during critical interaction phases. As illustrated in Fig. 3, a representative True Consensus session exhibits a high density of recurrent points ($RR = 0.747$). The prominent black block structures and diagonal segments indicate stable, shared physiological states. This structural resonance confirms that True Consensus is not merely a semantic agreement but a deep-seated state of physiological synchrony that our model successfully captures.

In contrast to the sprawling and chaotic patterns seen in Conflict or Neutral Discussion phases, the emergence of dense rectangular clusters (Laminarity) in Fig. 3 signifies that the dyad has entered a co-regulatory state. This physical evidence justifies the high attention weights assigned by our Dynamics-Informed Attention mechanism during such periods.

In addition, as presented in Table 5, the raw feature space exhibits a near-zero Silhouette score (0.051), reflecting the severe representation entanglement in high-frequency physiological streams. While independent late-fusion branches provide an initial boost in separability, the integration of Supervised Contrastive Loss in the CODEC framework yields a profound leap (+535.3% in Relative Gain). This non-linear optimization ensures that the network penetrates the majority-class dominance to learn robust, discriminative manifolds for rare cognitive states.

Table 5: Quantitative Evaluation of Latent Space Separability and Classification Performance

Feature Configuration	Accuracy	Silhouette Score	Relative Gain
Raw Early Fusion Inputs	81.9%	0.051	—
Late Fusion (w/o SupCon)	84.6%	0.193	+278.4%
Proposed CODEC (Full)	**86.5%**	**0.324**	**+535.3%**

Furthermore, visualizing the dynamic cross-modal attention weights reveals an interpretable physiological-behavioral alignment, as shown in Fig. 4. The inertial signals serve as a highly discriminative physical behavioral context for the electrophysiological arousal. Specifically, when the model detects anomalously steep EDA gradients accompanied by rigid, flat physical acceleration, the dynamics-informed attention mechanism successfully maps this localized stillness to high-intensity cognitive internalization and implicit psychological resistance.

6 Conclusion and Future Work

In this paper, we introduced the novel challenge of Multimodal Cognitive Synchrony Detection, moving beyond traditional superficial affective computing to uncover hidden interpersonal dynamics. To penetrate pervasive social masking phenomena such as Groupthink, we propose a dynamics-informed deep learning framework that integrates Subject-Aware Prompt Tuning, a CRQA dynamics prior injected into cross-modal attention, and Supervised Contrastive State Decoupling. Evaluated on our newly constructed CODEC dataset—which genuinely induces cognitive friction via the Hidden Profile Paradigm—our model achieved a remarkable progress with a weighted F1 of 0.85, outperforming state-of-the-art baselines in highly imbalanced, naturalistic scenarios.

Beyond algorithmic improvements, our study empirically establishes a physiological marker of agreement. We quantitatively reveal that high-quality True Consensus requires a preceding phase of physiological stress that is eventually resolved as structural resonance. In contrast, False Consensus is characterized by severe physiological decoupling and cognitive disengagement.

Future research will incorporate portable EEG to capture high-temporal-resolution cortical features and extend this dyadic framework to multi-party interactions. We will also focus on conducting field studies in real-world, high-stakes environments to validate the cross-domain robustness of the proposed framework, extending these physiological baselines to trained professionals. Ultimately, this work quantitatively verifies physiological synchrony as a consensus fingerprint and provides a solid data and technical foundation for intelligent decision-support systems combating groupthink. This advancement empowers future social-aware AI systems to monitor collaboration quality and intelligently intervene, thereby preventing catastrophic decision failures in high-stakes environments.

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